

Atlas V1 Finite Element Analysis

Upper Arm (Link 1) — Conservative Static Structural Assessment

Project Atlas | SolidWorks Simulation | August 2026

1. Objective

A finite element analysis (FEA) was performed on the Atlas V1 upper arm (Link 1) using SolidWorks Simulation. The purpose of the analysis was to evaluate the structural response of the 3D-printed component under representative static loading and identify regions of elevated stress and deformation.

The analysis focused on:

- Von Mises stress
- Resultant displacement
- Equivalent strain
- Critical stress locations
- Influence of the shoulder support condition

The analysis uses a simplified conservative support configuration in which the shoulder mounting bore is fully constrained. In the physical Atlas V1 assembly, the opposite side of the shoulder joint is additionally supported by a bearing. This bearing support was intentionally excluded from the FEA model to maintain a simplified conservative structural assessment.

2. Material Properties

The upper arm is designed to be manufactured from Anycubic PLA using fused-filament fabrication.

Material properties were obtained from the Anycubic PLA Technical Data Sheet, Version 3.0. The manufacturer reports properties based on printed test specimens and notes that mechanical performance depends on print orientation and processing conditions.

Property	Value
Density	1.24 g/cm ³
Young's Modulus (X-Y)	2534 ± 85 MPa
Tensile Strength (X-Y)	48 ± 5 MPa
Tensile Strength (Z)	28 ± 2.4 MPa
Elongation at Break (X-Y)	8 ± 1%
Bending Strength (X-Y)	90 ± 3 MPa
Bending Modulus (X-Y)	3360 ± 100 MPa

The SolidWorks Simulation model approximates the printed PLA component as a homogeneous material. Actual FDM-printed PLA exhibits direction-dependent mechanical behavior due to layer orientation and interlayer bonding. Therefore, the simulation results are treated as a first-order structural estimate rather than an exact prediction of printed-part failure.

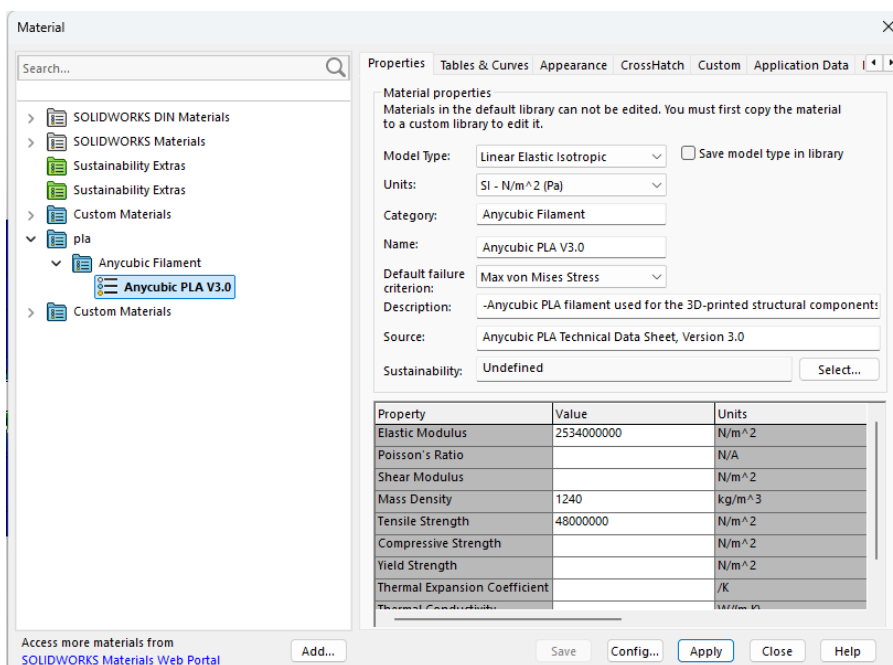


Figure 1: Anycubic PLA V3.0 Material Properties

3. Link 1 Static Structural Analysis

3.1 Analysis Configuration

A static structural study was created in SolidWorks Simulation to evaluate Link 1 under the loads transferred from the downstream arm components.

The analysis represents the loading condition developed from the previously completed Atlas V1 static shoulder torque analysis.

For the initial conservative study, the shoulder mounting bore was fully constrained. This simplifies the actual joint connection and intentionally does not account for the additional support provided by the bearing located on the opposite side of the assembled shoulder joint.

As a result, this study represents a simplified conservative structural model of Link 1. The actual assembly includes additional bearing support at the shoulder joint; therefore, the boundary conditions used in this study do not attempt to reproduce the complete joint contact behavior.

Applied Boundary Conditions

Boundary Condition	Value / Description
Study type	Static structural
Material	Anycubic PLA V3.0
Shoulder support	Fixed mounting bore
Gravity	9.81 m/s ²
Applied downstream force	2.71 N
Applied moment	0.221 N·m

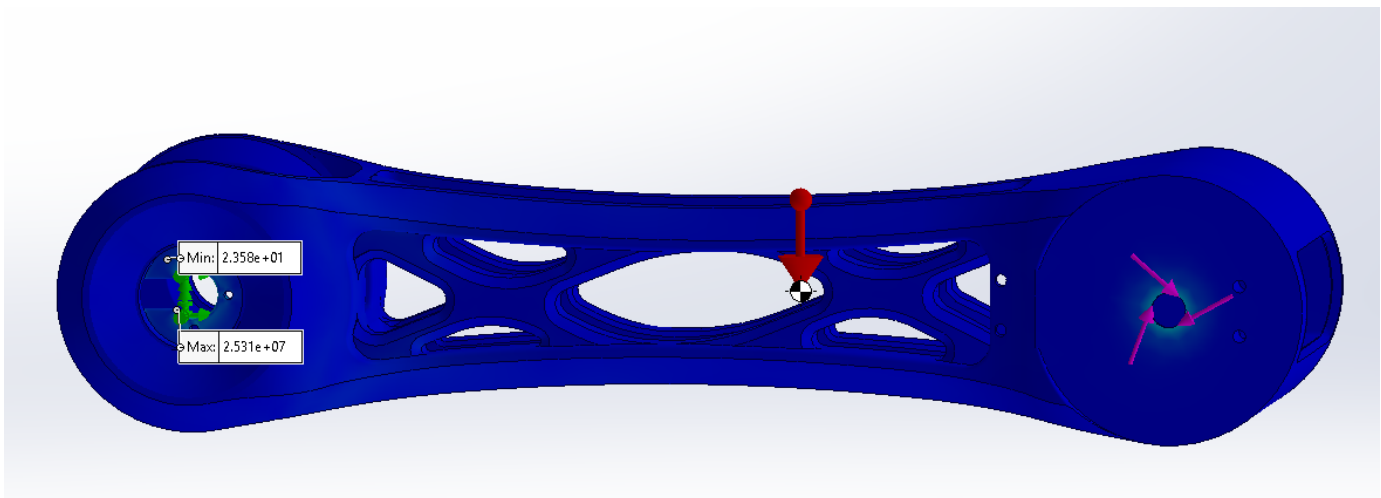


Figure 2: Conservative FEA Boundary Conditions

3.2 Mesh

A solid finite-element mesh was generated across the Link 1 geometry. The mesh captures the lightweight rib structure, joint interfaces, mounting holes, fillets, and other geometric features that may influence the stress distribution.

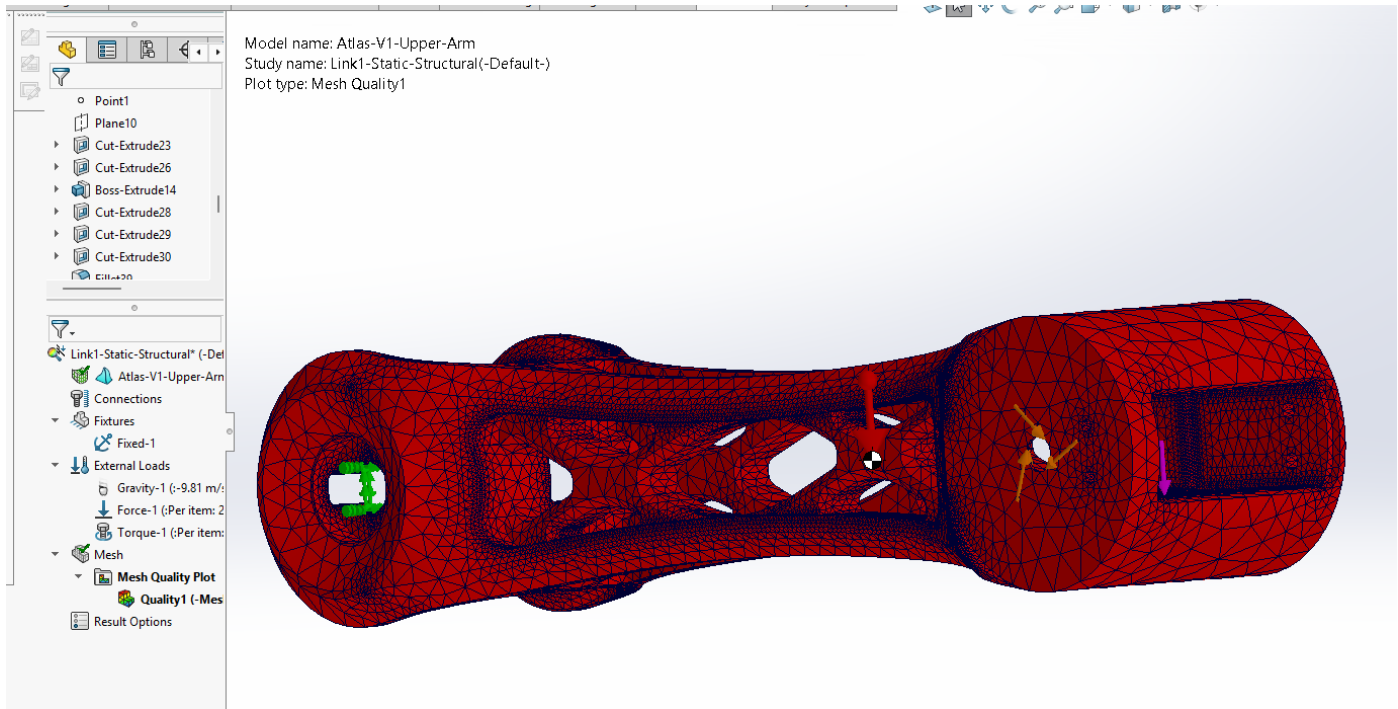


Figure 3: Link 1 Finite Element Mesh

A solid finite-element mesh was used to discretize the Link 1 geometry. The mesh was refined automatically around smaller geometric features, fillets, mounting holes, and the lightweight rib structure. The resulting mesh was considered sufficient for this preliminary structural assessment.

4. Conservative Analysis Results

4.1 Von Mises Stress

The static simulation predicted a maximum von Mises stress of **25.31 MPa**. The maximum stress occurs locally near the constrained shoulder mounting bore, where the applied loading is reacted through the fixed boundary condition. Most of the upper-arm structure experiences substantially lower stress, indicating that the predicted peak is localized around the shoulder connection rather than distributed throughout the lightweight rib structure.

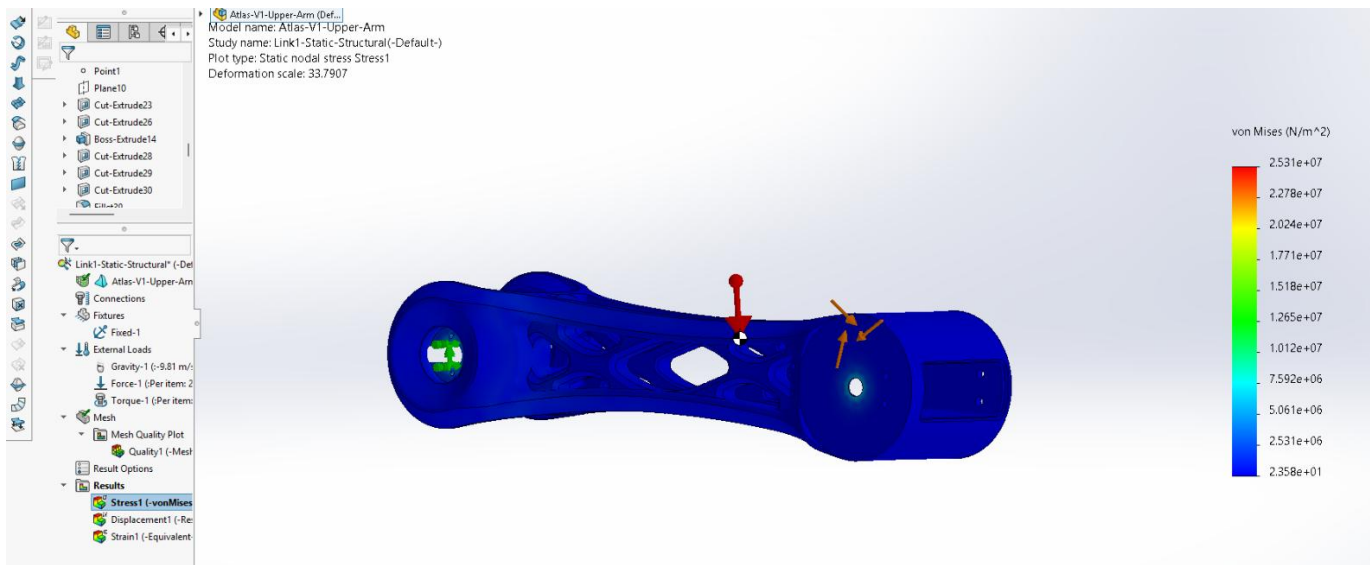


Figure 4: Von Mises Stress Distribution - Conservative Support Model

4.2 Resultant Displacement

The maximum resultant displacement predicted by the simulation was **0.755 mm**.

Maximum displacement occurs toward the unsupported distal end of Link 1, consistent with the component's expected bending response.

The deformation displayed in SolidWorks is visually exaggerated, with a deformation scale of approximately **33.79×**. Therefore, the displayed deformed shape should not be interpreted as the actual physical deformation magnitude.

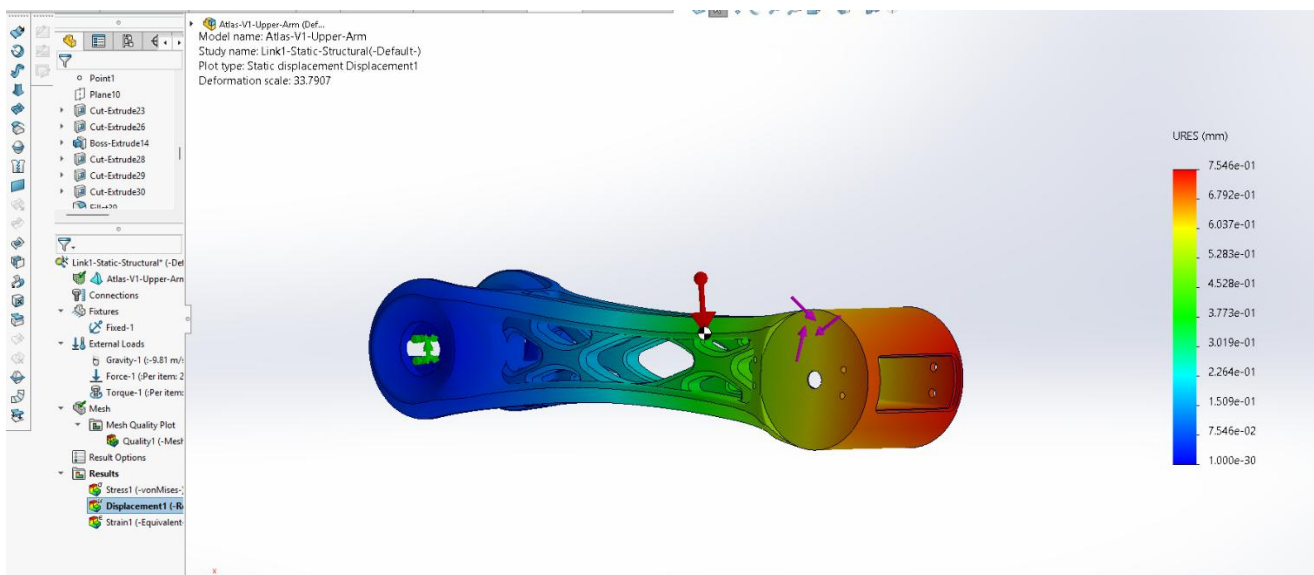


Figure 5: Resultant Displacement - Conservative Support Model

4.3 Equivalent Strain

The maximum equivalent strain predicted by the model was approximately **0.00450**.

The strain distribution follows the general stress response of the structure, with elevated values concentrated near the constrained joint region.

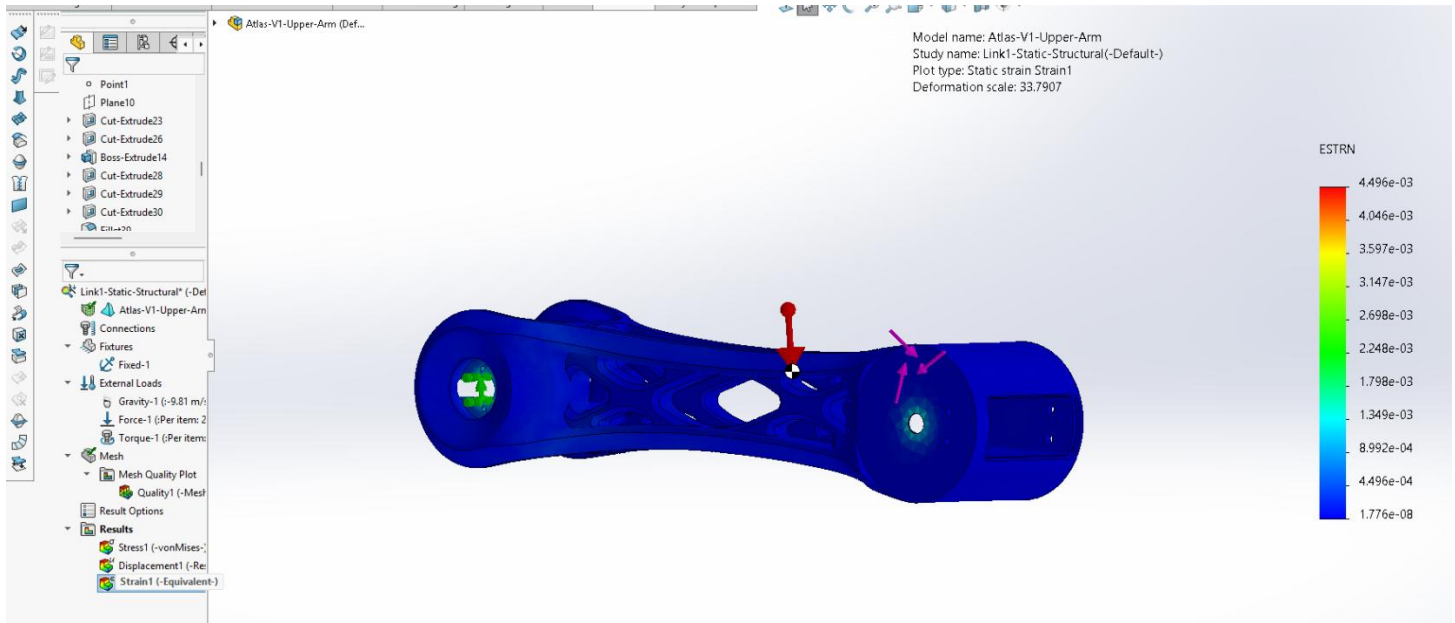


Figure 6: Equivalent Strain - Conservative Support Model

5. Conservative Analysis Summary

Result	Conservative Model
Maximum von Mises stress	25.31 MPa
Maximum resultant displacement	0.755 mm
Maximum equivalent strain	0.00450
Critical stress region	Shoulder mounting bore
X-Y tensile strength	48 ± 5 MPa
Nominal tensile-strength ratio	1.90
Z-direction tensile strength	28 ± 2.4 MPa
Nominal Z-direction strength ratio	1.11

5.1 Nominal Tensile-Strength-to-Stress Ratio

Using the manufacturer's typical X-Y tensile strength:

$$\text{X-Y Tensile Strength} = 48 \text{ MPa}$$

$$\text{Maximum von Mises Stress} = 25.31 \text{ MPa}$$

The nominal tensile-strength-to-stress ratio was calculated as:

$$\text{Strength Ratio} = \text{Tensile Strength} / \text{Maximum von Mises Stress}$$

$$\text{Strength Ratio} = 48 \text{ MPa} / 25.31 \text{ MPa}$$

$$\text{Strength Ratio} = 1.90$$

Therefore, the nominal X-Y tensile-strength-to-stress ratio for the conservative simulation is approximately **1.90**.

This value is not reported as a conventional yield-based factor of safety because the Anycubic PLA V3.0 technical data sheet does not provide a material yield strength. Additionally, the manufacturer's reported tensile properties are dependent on print orientation and processing conditions.

5.2 Z-Direction Strength Consideration

The manufacturer reports a lower Z-direction tensile strength of:

$$\text{Z-Direction Tensile Strength} = 28 \text{ MPa}$$

For comparison, the nominal Z-direction tensile-strength-to-stress ratio was calculated as:

$$\text{Strength Ratio (Z)} = \text{Z-Direction Tensile Strength} / \text{Maximum von Mises Stress}$$

$$\text{Strength Ratio (Z)} = 28 \text{ MPa} / 25.31 \text{ MPa}$$

$$\text{Strength Ratio (Z)} = 1.11$$

The resulting nominal Z-direction strength-to-stress ratio is approximately **1.11**.

This value should not be interpreted as a conventional factor of safety because the SolidWorks model approximates the printed PLA as a homogeneous isotropic material and does not explicitly model FDM layer orientation or interlayer bonding.

However, the relatively small Z-direction strength ratio indicates that print orientation is an important manufacturing consideration for Link 1. The component should therefore be printed in an orientation that minimizes critical structural loading across the weaker interlayer direction where practical.

6. Conclusion

A static finite element analysis was performed on the Atlas V1 upper arm (Link 1) to evaluate its structural response under representative loading derived from the Atlas V1 shoulder loading analysis.

The conservative model predicted a maximum von Mises stress of **25.31 MPa**, a maximum resultant displacement of **0.755 mm**, and a maximum equivalent strain of approximately **0.00450**. The highest predicted stress was localized near the constrained shoulder mounting bore, while the majority of the lightweight rib structure experienced substantially lower stress.

Comparison with the manufacturer's typical X-Y tensile strength of **48 MPa** produced a nominal tensile-strength-to-stress ratio of approximately **1.90**. Comparison with the manufacturer's typical Z-direction tensile strength of **28 MPa** produced a lower nominal ratio of approximately **1.11**, highlighting the importance of print orientation for the FDM-manufactured component.

The analysis contains several simplifying assumptions. The printed PLA was modeled as a homogeneous, linear-elastic, isotropic material, whereas actual FDM-printed PLA exhibits direction-dependent mechanical behavior. The shoulder mounting bore was also modeled as fully constrained, while the physical Atlas V1 shoulder assembly includes additional bearing support on the opposite side of the joint. Dynamic loading, impact, fastener preload, joint clearance, manufacturing defects, and detailed layer adhesion were not explicitly modeled.

Within these assumptions, the conservative static analysis indicates that the Link 1 geometry is suitable to proceed to **prototype fabrication and physical validation**. Final structural performance will be evaluated through testing of the manufactured Atlas V1 assembly.